NARAYANA IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | GTM-26 | Date: 25.01.2025

Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
B	B	B	C	A	A	B	D	A	C
11	12	13	14	15	16	17	18	19	20
D	D	C	B	D	B	A	C	D	B
21	22	23	24	25					
72	216	15	3	27					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	A	D	A	D	A	B	D	B	B
36	37	38	39	40	41	42	43	44	45
A	D	C	C	A	B	A	B	D	A
46	47	48	49	50					
2	1	5	3	3					

MATHEMATICS

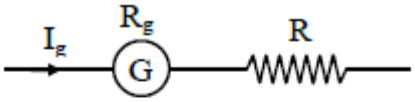
51	52	53	54	55	56	57	58	59	60
B	C	B	A	C	D	C	A	C	C
61	62	63	64	65	66	67	68	69	70
B	B	C	D	D	C	D	D	B	B
71	72	73	74	75					
14	169	678	72	46					

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PHYSICS

1.	Conceptual
2.	$g' = \frac{GMe}{(3R)^2} = \frac{1}{9}g$ $T = 2\pi\sqrt{\frac{\ell}{g'}}$ Since the time period of second pendulum is 2 sec. $T = 2$ sec $2 = 2\pi\sqrt{\frac{\ell}{g}9}$ $\boxed{\ell = \frac{1}{9}m}$
3.	$A = 80\ m^2$ Using Bernonlli equation $A(P_2 - P_1) = \frac{1}{2}\rho\left(V_1^2 - V_2^2\right)A$ $mg = \frac{1}{2} \times 1\ (70^2 - 50^2) \times 80$ $mg = 40 \times 2400$ $m = 9600$ kg
4.	$i = \frac{E_{eq}}{r_{eq}} = \frac{8 \times 5}{8 \times 0.2}$ $I = 25A$ $V = E - ir$ $= 5 - 0.2 \times 25$ $= 0$
5.	$V_C = \frac{q_{net}}{C_{net}} = \frac{CV + 2CV}{2C}$ $V_C = \frac{3V}{2}$ Loss of energy $= \frac{1}{2}CV^2 + \frac{1}{2}C(2V)^2 - \frac{1}{2}2C\left(\frac{3V}{2}\right)^2$ $= \left(\frac{CV^2}{4}\right)$

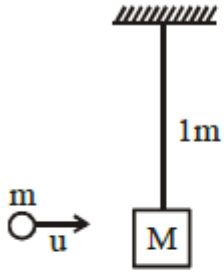
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6.	$C_V = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}$ $= \frac{2 \times \frac{3}{2}R + 6 \times \frac{5}{2}R}{2+6}$ $= \frac{9}{4}R$
7.	$T = m\omega^2 \ell$ $400 = 0.5\omega^2 \times 0.5$ $\omega = 40$ rad/s.
8.	$I = \frac{V}{X_C} = 230 \times 300 \times 200 \times 10^{-12} = 13.8\ \mu A$
9.	For PV^x = constant If work done by gas is asked then $W = \frac{nR\Delta T}{1-x}$ Here $x = \frac{3}{2}$ $\therefore W = \frac{P_2V_2 - P_1V_1}{-\frac{1}{2}}$ $= 2(P_1V_1 - P_2V_2)$ ….. Option (1) is correct If work done by external is asked then $W = -2(P_1V_1 - P_2V_2)$ ….. Option (2) is correct
10.	 $R = \frac{V}{I_g} - R_g = \frac{100}{5 \times 10^{-3}} - 50$ $= 20000 - 50$ $= 19950\Omega$
11.	$\lambda = \frac{h}{p} = \frac{h}{mv} \Rightarrow v = \frac{h}{m\lambda}$ $\frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} \times \frac{\lambda_\alpha}{\lambda_p}$ $= 4 \times 2 = 8$

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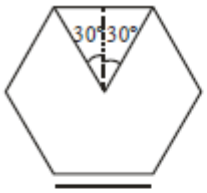
12.	$10\ \text{MSD} = 11\ \text{VSD}$ $1\ \text{VSD} = \frac{10}{11}\text{MSD}$ $\text{LC} = 1\text{MSD} - 1\text{VSD}$ $= 1\ \text{MSD} - \frac{10}{11}\text{MSD}$ $= \frac{1\text{MSD}}{11}$ $= \frac{5}{11}\ \text{units}$
13.	$\omega' = \omega$ $\frac{1}{\sqrt{L'C'}} = \frac{1}{\sqrt{LC}}$ $\therefore\ L'C' = LC$ $L'(4C) = LC$ $L' = \frac{L}{4}$ $\because$ Inductance must be decreased by $\frac{3L}{4}$
14.	$\rho = R\frac{\rho}{\ell}$ $\frac{\Delta\rho}{\rho} = \frac{\Delta R}{R} + 2\frac{\Delta r}{r} + \frac{\Delta\ell}{\ell}$ $= \frac{10}{100} + 2\times\frac{0.05}{0.35} + \frac{0.2}{15}$ $= \frac{1}{10} + \frac{2}{7} + \frac{1}{75}$ $\frac{\Delta\rho}{\rho} = 39.9\%$
15.	Angular impulse = change in angular momentum. [Angular impulse] = [Angular momentum] = [mvr] $= [M L^2 T^{-1}]$

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16.	 $mu = (M + m)V$ $10^{-2} \times 2 \times 10^{2} \cong 1 \times V$ $V \cong 2m/s$ $h = \frac{V^2}{2g} = 0.2\ m$
17.	$\frac{2\pi R}{T} = V$ Maximum height $H = \frac{v^2 \sin^2\theta}{2g}$ $4R = \frac{4\pi^2 R^2}{T^2 2g}\sin^2\theta$ $\sin\theta = \sqrt{\frac{2gT^2}{\pi^2 R}}$ $\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$
18.	$f_k = \mu N = 0.04 \times 20g = 8$ Newton $a = \frac{60-8}{26} = 2m/s^2$
19.	Transition from n = 1 to n = 3 $\Delta E = 12.1eV$
20.	Linear width $W = \frac{2\lambda d}{a} = \frac{2 \times 6 \times 10^{-7} \times 0.2}{1 \times 10^{-5}}$ $= 2.4 \times 10^{-2} = 24\ mm$

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21.



$\frac{4\pi}{6}$m

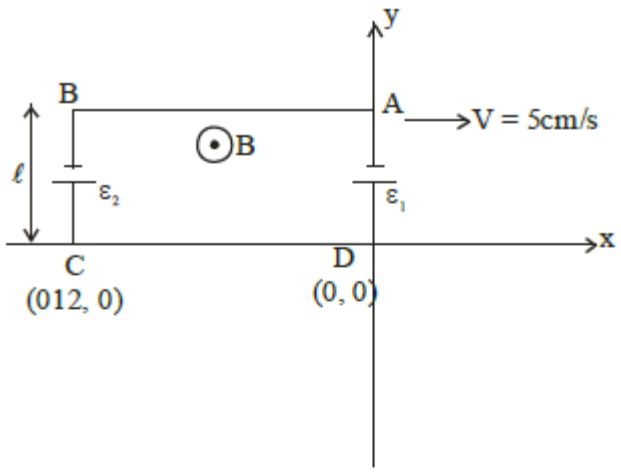
$$B = 6\left(\frac{\mu_0 I}{4\pi r}\right)(\sin 30^\circ + \sin 30^\circ)$$

$$= 6\frac{10^{-7} \times 4\pi\sqrt{3}}{\left(\frac{\sqrt{3} \times 4\pi}{2 \times 6}\right)}$$

$$= 72 \times 10^{-7} T$$

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22.



B_0 is the magnetic field at origin

$$\frac{dB}{dx} = -\frac{10^{-3}}{10^{-2}}$$

$$\int_{B_0}^{B} dB = -\int_0^x 10^{-1} dx$$

$$B - B_0 = -10^{-1}x$$

$$B = \left(B_0 - \frac{x}{10}\right)$$

Motional emf in AB = 0

Motional emf in CD = 0

Motional emf in AD = $\varepsilon_1 = B_0 \ell v$

Magnetic field on rod BC B

$$= \left(B_0 - \frac{(-12\times10^{-2})}{10}\right)$$

Motional emf in BC = $\varepsilon_2 = \left(B_0 + \frac{12\times10^{-2}}{10}\right)\ell \times v$

$\varepsilon_{eq} = \varepsilon_2 - \varepsilon_1 = 300 \times 10^{-7}$ V

For time variation

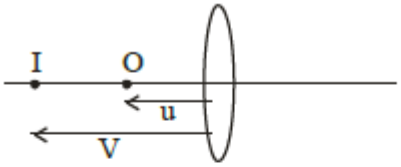
$$(\varepsilon_{eq})' = A\frac{dB}{dt} = 60 \times 10^{-7}\,V$$

$$(\varepsilon_{eq})_{net} = \varepsilon_{eq} + (\varepsilon_{eq})' = 360 \times 10^{-7}\,V$$

$$\text{Power} = \frac{(\varepsilon_{eq})_{net}^2}{R} = 216 \times 10^{-9}\,W$$

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23.



$v = 3u$

$v - u = 20$ cm

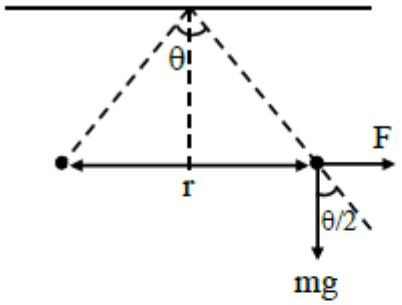
$2u = 20$ cm

$u = 10$ cm

$$\frac{1}{(-30)} - \frac{1}{(-10)} = \frac{1}{f}$$

$f = 15$ cm

24.



In air $\tan\frac{\theta}{2} = \frac{F}{mg} = \frac{q^2}{4\pi\varepsilon_0 r^2 mg}$

In water $\tan\frac{\theta}{2} = \frac{F'}{mg'} = \frac{q^2}{4\pi\varepsilon_0\varepsilon_r r^2 mg_{eff}}$

Equate both equations

$$\varepsilon_0 g = \varepsilon_0\, \varepsilon_r\, g \left[1 - \frac{1}{1.5}\right]$$

$\varepsilon_r = 3$

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25.

$$R = R_0 A^{1/3}$$

$$R^3 \propto A$$

$$\left(\frac{4.8}{4}\right)^3 = \frac{64}{A}$$

$$= \frac{64}{A} = (1.2)^3$$

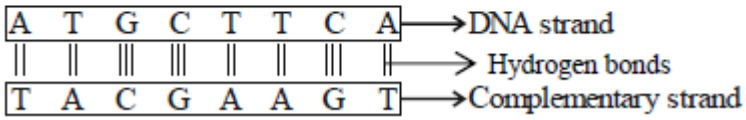
$$\frac{64}{A} = 1.44 \times 1.2$$

$$A = \frac{64}{1.44 \times 1.2} = \frac{1000}{x}$$

$$x = \frac{144 \times 12}{64} = 27$$

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26. **Sol.** Adenine base pairs with thymine with 2 hydrogen bonds and cytosine base pairs with guanine with 3 hydrogen bonds.



27. (i) $KCN + R-X \rightarrow R-CN$ (Major product) (Ionic)

(ii) $AgCN + R-X \rightarrow R-NC$ (Major product) (Covalent)

AgCN is mainly covalent in nature and nitrogen is available for attack, so alkyl isocyanide is formed as main product.

28. The balanced reaction is,

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

X = 14

Y = 6

A = 7

29. When a particular oxidation state becomes less stable relative to other oxidation state, one lower, one higher, it is said to undergo disproportionation.

$Cu^+ \rightarrow Cu^{2+} + Cu$

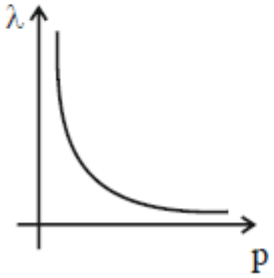
$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

30. In F^-, Ne, Na^+ all have $1s^2$, $2s^2$, $2p^6$ configuration. They have different size due to the difference in nuclear charge.

31. $\lambda = \frac{h}{p}\left[\lambda \propto \frac{1}{p}\right]$

$\Rightarrow \lambda p = h$ (constant)

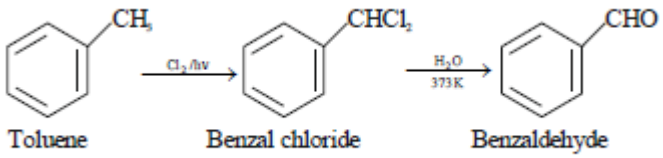
So, the plot is a rectangular hyperbola.



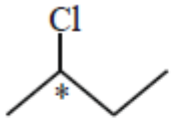
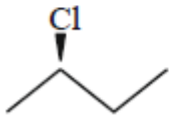
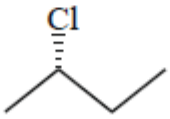
32. $[Ni(H_2O)_6]^{+2}$ → Green colour solution due to d-d transition.

$[Ni(CN)_4]^{-2}$ → is diamagnetic and it is colourless.

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33.	Unlike NH_3, PH_3 molecules are not associated through hydrogen bonding in liquid state. That is why the boiling point of PH_3 is lower than NH_3.
34.	 Toluene Benzal chloride Benzaldehyde
35.	Aniline is also known as amino benzene.
36.	In Homoleptic complex all the ligand attached with the central atom should be the same. Hence $[Ni(CN)_4]^{2-}$ is a homoleptic complex.
37.	Higher the electron density in the benzene ring more easily it will be attacked by an electrophile. Phenol has the highest electron density amongst all the given compound.
38.	Heterolytic cleavage of Bond lead to formation of ions.
39.	Increasing order of ionic character $N_2 < SO_2 < ClF_3 < K_2O < LiF$ Ionic character depends upon difference of electronegativity (bond polarity).
40.	<table><tr><td rowspan="2">Salt</td><td colspan="3">Values of i (for different conc. of a Salt)</td></tr><tr><td>0.1 M</td><td>0.01 M</td><td>0.001 M</td></tr><tr><td>NaCl</td><td>1.87</td><td>1.94</td><td>1.94</td></tr></table> i approach 2 as the solution become very dilute.
41.	Kjeldahl's method is used for estimation of Nitrogen where $CuSO_4$ acts as a catalyst.
42.	**Statement (I)** : Potassium hydrogen phthalate is a primary standard for standardisation of sodium hydroxide solution as it is economical and its concentration does not changes with time. Phenophthalin can acts as indicator in acid base titration as it shows colour in pH range 8.3 to 10.1
43.	$CH_3(CH_2)_5COOC_2H_5 \xrightarrow{DIBAL-H,\ H_2O} CH_3(CH_2)_5CHO$ $C_6H_5COC_6H_5 \xrightarrow{Zn(Hg)\ \&\ conc.\ HCl} C_6H_5CH_2C_6H_5$ $C_6H_5CHO \xrightarrow[H_2O]{CH_3MgBr} C_6H_5CH(OH)CH_3$ $CH_3COCH_2COOC_2H_5 \xrightarrow{NaBH_4,\ H^+} CH_3CH(OH)CH_2COOC_2H_5$

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44.	During free expansion of an ideal gas under adiabatic condition $q = 0$, $\Delta T = 0$, $w = 0$.
45.	The NH_2 group in Aniline is ortho and para directing and a powerful activating group as NH_2 has strong +M effect. Aniline does not undergo Friedel-Craft's reaction (alkylation and acylation) as Aniline will form complex with $AlCl_3$ which will deactivate the benzene ring.
46.	 There is one chiral centre present in given compound. So, Total optical isomers = 2  and 
47.	$E = E^{\circ}_{H^+/H_2} - \frac{0.06}{2}\log\frac{P_{H_2}}{[H^+]^2}$ $E = 0.00 - \frac{0.06}{2}\log\frac{2}{[1]^2}$ $E = -0.03 \times 0.3 = -0.9 \times 10^{-2}\,V$

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48.	$SrSO_4$ – white $Mg(NH_4)PO_4$ – white $BaCrO_4$ – yellow $Mn(OH)_2$ – white $PbSO_4$ – white $PbCrO_4$ – yellow AgBr – pale yellow PbI_2 – yellow CaC_2O_4 – white $[Fe(OH)_2(CH_3COO)]$ – Brown Red
49.	Acidic oxide: Cl_2O_7, SiO_2, N_2O_5 Neutral oxide: CO, NO, N_2O Amphoteric oxide: Al_2O_3 , SnO_2 , PbO_2
50.	PF_5, PCl_5, $Fe(CO)_5$; Trigonal bipyramidal BrF_5 ; square pyramidal $[PtCl_4]^{-2}$; square planar BF_3 ; Trigonal planar

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51.

Sol.

$$P(4W4B/2W2B) = \frac{P(4W4B)\times P(2W2B/4W4B)}{P(2W6B)\times P(2W2B/2W6B)+P(3W5B)\times P(2W2B/3W5B)+\ldots\ldots\ldots+P(6W2B)\times P(2W2B/6W2B)}$$

$$= \frac{\frac{1}{5}\times\frac{{}^4C_2\times{}^4C_2}{{}^8C_4}}{\frac{1}{5}\times\frac{{}^2C_2\times{}^6C_2}{{}^8C_4}+\frac{1}{5}\times\frac{{}^3C_2\times{}^5C_2}{{}^8C_4}+\ldots+\frac{1}{5}\times\frac{{}^6C_2\times{}^2C_2}{{}^8C_4}}$$

$$= \frac{2}{7}$$

52.

$$\int_0^{\frac{\pi}{4}}\frac{xdx}{\sin^4(2x)+\cos^4(2x)}$$

Let $2x=t$ then $dx=\frac{1}{2}dt$

$$I=\frac{1}{4}\int_0^{\frac{\pi}{2}}\frac{tdt}{\sin^4 t+\cos^4 t}$$

$$I=\frac{1}{4}\int_0^{\frac{\pi}{2}}\frac{\left(\frac{\pi}{2}-t\right)dt}{\sin^4\left(\frac{\pi}{2}-t\right)+\cos^4\left(\frac{\pi}{2}-t\right)}$$

$$I=\frac{1}{4}\int_0^{\frac{\pi}{2}}\frac{\frac{\pi}{2}dt}{\sin^4 t+\cos^4 t}-I$$

$$2I=\frac{\pi}{8}\int_0^{\frac{\pi}{2}}\frac{dt}{\sin^4 t+\cos^4 t}$$

$$2I=\frac{\pi}{8}\int_0^{\frac{\pi}{2}}\frac{\sec^4 tdt}{\tan^4 t+1}$$

Let tant = y then $\sec^2 t\,dt = dy$

$$2I=\frac{\pi}{8}\int_0^{\infty}\frac{(1+y^2)dy}{1+y^4}$$

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$$= \frac{\pi}{16}\int_0^{\infty} \frac{1+\frac{1}{y^2}}{y^2+\frac{1}{y^2}}dy$$

Put $y-\frac{1}{y}=p$

$$I = \frac{\pi}{16}\int_{-\infty}^{\infty} \frac{dp}{p^2+\left(\sqrt{2}\right)^2}$$

$$= \frac{\pi}{16\sqrt{2}}\left[\tan^{-1}\left(\frac{p}{\sqrt{2}}\right)\right]_{-\infty}^{\infty}$$

$$I = \frac{\pi^2}{16\sqrt{2}}$$

53.

$$A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix} \Rightarrow \det(A) = 3$$

$$B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow \det(B) = 1$$

Now $C = ABA^T \Rightarrow \det(C) = (\det(A))^2 \times \det(B)$

$|C| = 9$

Now $|X| = |A^T C^2 A|$

$= |A^T| \, |C|^2 \, |A|$

$= |A|^2 \, |C|^2$

$= 9 \times 81$

$= 729$

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54.	Finding tan (A + B) we get $\Rightarrow \tan(A+B) = \dfrac{\tan A+\tan B}{1-\tan A\tan B} = \dfrac{\dfrac{1}{\sqrt{x(x^2+x+1)}}+\dfrac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1-\dfrac{1}{x^2+x+1}}$ $\Rightarrow \tan(A+B) = \dfrac{(1+x)\left(\sqrt{x^2+x+1}\right)}{\left(x^2+x\right)\left(\sqrt{x}\right)}$ $\dfrac{(1+x)\left(\sqrt{x^2+x+1}\right)}{\left(x^2+x\right)\left(\sqrt{x}\right)}$ $\tan(A+B) = \dfrac{\sqrt{x^2+x+1}}{x\sqrt{x}} = \tan C$ $A+B=C$
55.	Total ways to partition 5 into 4 parts are : 5, 0, 0, 0 $\Rightarrow$ 1 way 4, 1, 0, 0 $\Rightarrow \dfrac{5!}{4!} = 5$ ways 3, 2, 0, 0, $\Rightarrow \dfrac{5!}{3!2!} = 10$ ways $2,2,0,1 \Rightarrow \dfrac{5!}{2!2!2!} = 15$ ways $2,1,1,1 \Rightarrow \dfrac{5!}{2!(1!)^3 3!} = 10$ ways $3,1,1,0 \Rightarrow \dfrac{5!}{3!2!} = 10$ ways Total $\Rightarrow$ 1+5+10+15+10+10 = 51 ways

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56.

Let $Z = x + iy$

Then $(x - 1)^2 + y^2 = 1 \quad \rightarrow (1)$

& $\left(\sqrt{2}-1\right)(2x) - i(2iy) = 2\sqrt{2}$

$\Rightarrow (\sqrt{2}-1)x + y = \sqrt{2} \rightarrow (2)$

Solving (1) & (2) we get

Either $x = 1$ or $x = \dfrac{1}{2-\sqrt{2}} \rightarrow (3)$

On solving (3) with (2) we get

For $x = 1 \Rightarrow y = 1 \Rightarrow Z_2 = 1 + i$

& for

$$x = \frac{1}{2-\sqrt{2}} \Rightarrow y = \sqrt{2} - \frac{1}{\sqrt{2}} \Rightarrow Z_1 = \left(1 + \frac{1}{\sqrt{2}}\right) + \frac{i}{\sqrt{2}}$$

Now

$$\left|\sqrt{2}z_1 - z_2\right|^2$$

$$= \left|\left(\frac{1}{\sqrt{2}} + 1\right)\sqrt{2} + i - (1+i)\right|^2$$

$$= \left(\sqrt{2}\right)^2$$

$$= 2$$

57.

Median $= 170 \Rightarrow 125, a, b, 170, 190, 210, 230$

Mean deviation about

Median $=$

$$\frac{0+45+60+20+40+170-a+170-b}{7} = \frac{205}{7}$$

$\Rightarrow a + b = 300$

Mean $= \dfrac{170+125+230+190+210+a+b}{7} = 175$

Mean deviation

About mean $=$

$$\frac{50+175-a+175-b+5+15+35+55}{7} = 30$$

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58.

$\vec{a} = -5\hat{i} + j - 3\hat{k}$

$\vec{b} = \hat{i} + 2\hat{j} - 4\hat{k}$

$(\vec{a} \times \vec{b}) \times \hat{i} = (\vec{a} \cdot \hat{i})\vec{b} - (\vec{b} \cdot \hat{i})\vec{a}$

$= -5\vec{b} - \vec{a}$

$= \left(\left(\left(-5\vec{b} - \vec{a}\right) \times \hat{i}\right) \times \hat{i}\right)$

$= \left(\left(-11\hat{j} + 23\hat{k}\right) \times \hat{i}\right) \times \hat{i}$

$\Rightarrow \left(11\hat{k} + 23\hat{j}\right) \times \hat{i}$

$\Rightarrow \left(11\hat{j} - 23\hat{k}\right)$

$\vec{c}.\left(-\hat{i} + \hat{j} + \hat{k}\right) = 11 - 23 = -12$

59.

$\left(\sqrt{3} + \sqrt{2}\right)^x + \left(\sqrt{3} - \sqrt{2}\right)^x = 10$

Let $\left(\sqrt{3} + \sqrt{2}\right)^x = t$

$t + \frac{1}{t} = 10$

$t^2 - 10t + 1 = 0$

$t = \frac{10 \pm \sqrt{100 - 4}}{2} = 5 \pm 2\sqrt{6}$

$\left(\sqrt{3} + \sqrt{2}\right)^x = \left(\sqrt{3} \pm \sqrt{2}\right)^2$

$x = 2$ or $x = -2$

Number of solutions = 2

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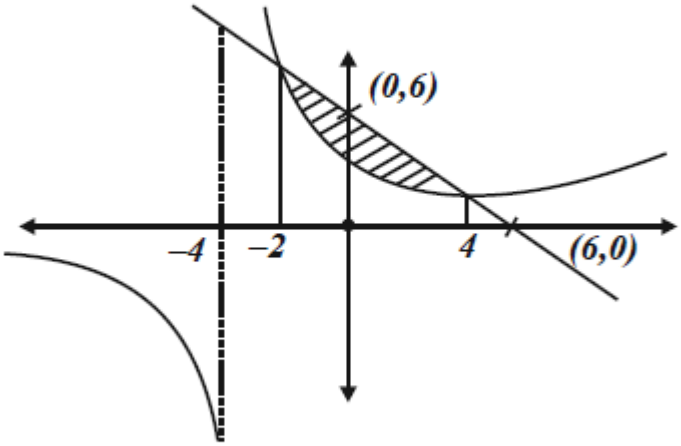
60.

$xy + 4y = 16$, $x + y = 6$

$y(x + 4) = 16$ ____(1) , $x + y = 6$___(2)

on solving, (1) & (2)

we get $x = 4, x = -2$



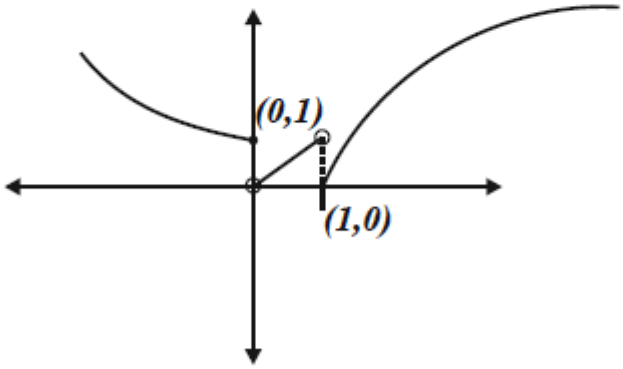
$$\text{Area} = \int_{-2}^{4}\left((6-x)-\left(\frac{16}{x+4}\right)\right)dx$$

$$= 30 - 32\ln 2$$

61.

$$g(f(x)) = \begin{cases} f(x), f(x) \geq 0 \\ e^{f(x)}, f(x) < 0 \end{cases}$$

$$g(f(x)) = \begin{cases} e^{-x}, (-\infty, 0] \\ e^{\ln x}, (0,1) \\ \ln x, [1, \infty) \end{cases}$$



Graph of g(f(x))

g(f(x)) $\Rightarrow$ Many one into

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62.	Using family of planes $2x + 3y - z - 5 = k_1 (x + \alpha y + 3z + 4) + k_2 (3x - y + \beta z - 7)$ $2 = k_1 + 3k_2,\ 3 = k_1\alpha - k_2,\ -1 = 3k_1 + \beta k_2,\ -5 = 4k_1 - 7k_2$ On solving we get $k_2 = \frac{13}{19}, k_1 = \frac{-1}{19}, \alpha = -70, \beta = \frac{-16}{13}$ $13\alpha\beta = 13(-70)\left(\frac{-16}{13}\right)$ $= 1120$
63.	$e_h = \sqrt{1+\sin^2\theta}$ $e_c = \sqrt{1-\sin^2\theta}$ $e_h = \sqrt{7}e_c$ $1+\sin^2\theta = 7(1-\sin^2\theta)$ $\sin^2\theta = \frac{6}{8} = \frac{3}{4}$ $\sin\theta = \frac{\sqrt{3}}{2}$ $\theta = \frac{\pi}{3}$

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64.

$$\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$$

$$x + y = t$$

$$\frac{dt}{dx} - 1 = 2xt^3 - xt - 1$$

$$\frac{dt}{2t^3 - t} = xdx$$

$$\frac{\mathrm{tdt}}{2\mathrm{t}^4 - \mathrm{t}^2} = \mathrm{xdx}$$

Let $t^2 = z$

$$\int \frac{\mathrm{dz}}{2\left(2\mathrm{z}^2 - \mathrm{z}\right)} = \int \mathrm{xdx}$$

$$\int \frac{dz}{4z\left(z - \frac{1}{2}\right)} = \int xdx$$

$$\ln\left|\frac{z - \frac{1}{2}}{z}\right| = x^2 + k$$

$$z = \frac{1}{2 - \sqrt{e}}$$

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65.

At x = 1, f(x) is continuous therefore,

$f(1^-) = f(1) = f(1^+)$

$f(1) = 3 + c$ ....(1)

$f(1^+) = \lim_{h\to 0} 2(1 + h) + 1$

$f(1^+) = \lim_{h\to 0} 3 + 2h = 3$ ....(2)

from (1) & (2)

$c = 0$

at x = 0, f(x) is continuous therefore,

$f(0^-) = f(0) = f(0^+)$ ....(3)

$f(0) = f(0^+) = 2$ ....(4)

$f(0^-)$ has to be equal to 2

$$\lim_{h\to 0}\frac{a - b\cos(2h)}{h^2}$$

$$\lim_{h\to 0}\frac{a - b\left\{1 - \frac{4h^2}{2!} + \frac{16h^4}{4!} + \ldots\right\}}{h^2}$$

$$\lim_{h\to 0}\frac{a - b + b\left\{2h^2 - \frac{2}{3}h^4 \ldots\right\}}{h^2}$$

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	for limit to exist a – b = 0 and limit is 2b ….(5) from (3), (4) & (5) a = b = 1 checking differentiability at x = 0 LHD : $\lim_{h\to 0}\dfrac{\dfrac{1-\cos 2h}{h^2}-2}{-h}$ $\lim_{h\to 0}\dfrac{1-\left(1-\dfrac{4h^2}{2!}+\dfrac{16h^4}{4!}\ldots\right)-2h^2}{-h^3}=0$ RHD : $\lim_{h\to 0}\dfrac{(0+h)^2+2-2}{h}=0$ Function is differentiable at every point in its domain $\therefore$ m = 0 m + a + b + c = 0 + 1 + 1 + 0 = 2
66.	$e=\dfrac{1}{\sqrt{2}}=\sqrt{1-\dfrac{b^2}{a^2}}\Rightarrow\dfrac{1}{2}=1-\dfrac{b^2}{a^2}$ $\dfrac{2b^2}{a}=14$ $e_H=\sqrt{1+\dfrac{b^2}{a^2}}=\sqrt{1+\dfrac{1}{2}}=\sqrt{\dfrac{3}{2}}$ $(e_H)^2=\dfrac{3}{2}$
67.	3, a, b, c → A.P $\Rightarrow$ 3, 3+d, 3+2d, 3+3d 3, a–1, b+1, c+9 → G.P $\Rightarrow$ 3, 2+d, 4+2d, 12+3d a = 3 + d $(2+d)^2=3(4+2d)$ b = 3 + 2d d = 4, -2 c = 3 + 3d If d = 4 G.P $\Rightarrow$ 3, 6, 12, 24 a = 7 b = 11 c = 15 $\dfrac{a+b+c}{3}=11$

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68.

$x^2 + y^2 = 4$

C (0, 0) $\qquad r_1 = 2$

C' $(2\lambda, 0)$ $\quad r_2 = \sqrt{4\lambda^2 - 9}$

$|r_1 - r_2| < CC' < |r_1 + r_2|$

$$\left|2 - \sqrt{4\lambda^2 - 9}\right| < |2\lambda| < 2 + \sqrt{4\lambda^2 - 9}$$

$$4 + 4\lambda^2 - 9 - 4\sqrt{4\lambda^2 - 9} < 4\lambda^2$$

True $\lambda \in R$.... (1)

$$4\lambda^2 < 4 + 4\lambda^2 - 9 + 4\sqrt{4\lambda^2 - 9}$$

$$5 < 4\sqrt{4\lambda^2 - 9} \text{ and } \qquad \lambda^2 \geq \frac{9}{4}$$

$$\frac{25}{16} < 4\lambda^2 - 9 \qquad \lambda \in \left(-\infty, -\frac{3}{2}\right] \cup \left[\frac{3}{2}, \infty\right)$$

$$\frac{169}{64} < \lambda^2$$

$$\lambda \in \left(-\infty, -\frac{13}{8}\right) \cup \left(\frac{13}{8}, \infty\right) \qquad \ldots(2)$$

from (1) and (2) $\lambda \in$

$$\lambda \in \left(-\infty, -\frac{13}{8}\right) \cup \left(\frac{13}{8}, \infty\right) \Rightarrow R - \left[-\frac{13}{8}, \frac{13}{8}\right]$$

as per question $a = -\frac{13}{8}$ and $b = \frac{13}{8}$

required point is (–1, 6) with satisfies option (4)

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69.

$5\,f(x)+4\,f\left(\frac{1}{x}\right)=x^2-2,\ \forall x\neq 0\ldots(1)$

Substitute $x\to\frac{1}{x}$

$5f\left(\frac{1}{x}\right)+4f(x)=\frac{1}{x^2}-2\quad\ldots(2)$

On solving (1) and (2)

$f(x)=\frac{5x^4-2x^2-4}{9x^2}$

$y=9x^2f(x)$

$y=5x^4-2x^2-4\quad\ldots(3)$

$\frac{dy}{dx}=20x^3-4x$

for strictly increasing

$\frac{dy}{dx}>0$

$4x(5x^2-1)>0$

$x\in\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

70.

Passing points of lines L_1 & L_2 are

$(\lambda,2,1)\ \&\ \left(\sqrt{3},1,2\right)$

$$S.D=\frac{\begin{vmatrix}\sqrt{3}-\lambda & -1 & 1\\ -2 & 1 & 1\\ 1 & -2 & 1\end{vmatrix}}{\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -2 & 1 & 1\\ 1 & -2 & 1\end{vmatrix}}$$

$$1=\left|\frac{\sqrt{3}-\lambda}{\sqrt{3}}\right|$$

$\lambda=0,\lambda=2\sqrt{3}$

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71.

$(t+1)dx = (2x + (t+1)^4)dt$

$$\frac{dx}{dt} = \frac{2x + (t+1)^4}{t+1}$$

$$\frac{dx}{dt} - \frac{2x}{t+1} = (t+1)^3$$

$$I.F = e^{-\int \frac{2}{t+1} dt} = e^{-2\ln(t+1)} = \frac{1}{(t+1)^2}$$

$$\frac{x}{(t+1)^2} = \int \frac{1}{(t+1)^2}(t+1)^3 dt + c$$

$$\frac{x}{(t+1)^2} = \frac{(t+1)^2}{2} + c$$

$$\Rightarrow c = \frac{3}{2}$$

$$x = \frac{(t+1)^4}{2} + \frac{3}{2}(t+1)^2$$

put, $t = 1$

$x = 2^3 + 6 = 14$

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72.

$x + 2y + 3z = 42, \quad x, y, z \geq 0$

$z = 0 \quad x + 2y = 42 \Rightarrow 22$

$z = 1 \quad x + 2y = 39 \Rightarrow 20$

$z = 2 \quad x + 2y = 36 \Rightarrow 19$

$z = 3 \quad x + 2y = 33 \Rightarrow 17$

$z = 4 \quad x + 2y = 30 \Rightarrow 16$

$z = 5 \quad x + 2y = 27 \Rightarrow 14$

$z = 6 \quad x + 2y = 24 \Rightarrow 13$

$z = 7 \quad x + 2y = 21 \Rightarrow 11$

$z = 8 \quad x + 2y = 18 \Rightarrow 10$

$z = 9 \quad x + 2y = 15 \Rightarrow 8$

$z = 10 \quad x + 2y = 12 \Rightarrow 7$

$z = 11 \quad x + 2y = 9 \Rightarrow 5$

$z = 12 \quad x + 2y = 6 \Rightarrow 4$

$z = 13 \quad x + 2y = 3 \Rightarrow 2$

$z = 14 \quad x + 2y = 0 \Rightarrow 1$

Total : 169

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73.

coeff of x^{30} in $\dfrac{(x+1)^6(1+x^2)^7(1-x^3)^8}{x^6}$

coeff. of x^{36} in $(1+x)^6(1+x^2)^7(1-x^3)^8$

General term

$${}^6C_{r_1}\,{}^7C_{r_2}\,{}^8C_{r_3}(-1)^{r_3}x^{r_1+2r_2+3r_3}$$

$r_1+2r_2+3r_3=36$

Case-I :

r_1	r_2	r_3
0	6	8
2	5	8
4	4	8
6	3	8

$r_1+2r_2=12$ (Taking $r_3=8$)

Case-II :

r_1	r_2	r_3
1	7	7
3	6	7
5	5	7

$r_1+2r_2=15$ (Taking $r_3=7$)

Case-III :

r_1	r_2	r_3
4	7	6
6	6	6

$r_1+2r_2=18$ (Taking $r_3=6$)

Coeff. $= 7+(15\times21)+(15\times35)+(35)$

$-(6\times8)-(20\times7\times8)-(6\times21\times8)+(15\times28)$

$+(7\times28)=-678=\alpha$

$|\alpha|=678$

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74.

$x^2 + y^2 = 3$ and $x^2 = 2y$

$y^2 + 2y - 3 = 0 \Rightarrow (y + 3)(y - 1) = 0$

$y = -3$ or $y = 1$

$y = 1$ $x = \sqrt{2} \Rightarrow P(\sqrt{2}, 1)$

p lies on the line

$\sqrt{2}x + y = \alpha$

$\sqrt{2}(\sqrt{2}) + 1 = \alpha$

$\alpha = 3$

For circle C_1

Q_1 lies on y axis

Let Q_1 $(0, \alpha)$ coordinates

$R_1 = 2\sqrt{3}$ (Given

Line L act as tangent

Apply P = r (condition of tangency)

$\Rightarrow \left|\frac{\alpha - 3}{\sqrt{3}}\right| = 2\sqrt{3}$

$\Rightarrow |\alpha - 3| = 6$

$\alpha - 3 = 6$ or $\alpha - 3 = -6$

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	$\Rightarrow \left\lvert \frac{\alpha-3}{\sqrt{3}} \right\rvert = 2\sqrt{3}$ $\Rightarrow \lvert \alpha-3 \rvert = 6$ $\alpha-3=6$ or $\alpha-3=-6$ $\Rightarrow \alpha=9$ $\alpha=-3$ $\triangle PQ_1Q_2 = \frac{1}{2}\begin{vmatrix} \sqrt{2} & 1 & 1 \\ 0 & 9 & 1 \\ 0 & -3 & 1 \end{vmatrix}$ $= \frac{1}{2}\left(\sqrt{2}(12)\right) = 6\sqrt{2}$ $\left(\triangle PQ_1Q_2\right)^2 = 72$
75.	$n(R_1) = 20+10+6+5+4+3+2+2+2$ $+2+\underbrace{1+\ldots+1}_{10\text{ times}}$ $n(R_1) = 66$ $R_1 \cap R_2 = \{(1,1),(2,2),\ldots(20,20)\}$ $n(R_1 \cap R_2) = 20$ $n(R_1 - R_2) = n(R_1) - n(R_1 \cap R_2)$ $= n(R_1) - 20$ $= 66 - 20$ $R_1 - R_2 = 46$ Pair

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